Brownian dynamics of nanoparticles in optical tweezers

Loading and escape times of a focused Gaussian-beam trap

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Abstract

A nanoparticle held in an optical tweezer is studied as an overdamped Brownian particle in the gradient-force potential of a focused Gaussian beam, and two timescales are computed. The first is how long the trap takes to load from a uniform initial distribution, and the second is how long a trapped particle survives before thermal fluctuations carry it out. Both are obtained from a Chang-Cooper finite-volume solution of the Fokker-Planck equation and, independently, from ensembles of Langevin trajectories. A one-dimensional reduction of the trap provides an analytic anchor, because there the mean escape time is known in closed form, and five independent routes agree to better than 0.1% at a depth of $8\,k_{\mathrm{B}}T$. The full three-dimensional trap is solved on the (ρ, z) half-plane by exploiting axisymmetry, and validated against a Cartesian three-dimensional particle ensemble. Loading is found to be governed by diffusion across the surrounding domain rather than by the local dynamics of the well, and in three dimensions it proceeds in two stages, so a fit-free integral relaxation time is used in place of a single exponential time constant. Escape is exponential and dominated by a single slow mode. A constant axial force is superimposed on the trap as a conservative stand-in for a real measured force field, tilting the well while preserving the Boltzmann steady state that the validation relies on, which establishes the cross-validation framework needed for the non-conservative scattering force that a real trap exerts.

1 Introduction

An optical tweezer traps a small particle near the focus of a tightly focused laser beam [1, 2, 3]. The electric field produced by the beam polarises the particle, and because the field is strongest at the focus, the particle experiences a force that pulls it up the intensity gradient towards the trap. Immersed in water, it moves within a smooth potential well while continual collisions with solvent molecules drive random thermal motion. Since the particle is very small and the surrounding fluid is highly viscous, any momentum it acquires is dissipated almost immediately. Its velocity therefore adjusts to changes in the applied forces much more rapidly than its position changes and can be treated as remaining in equilibrium with the current position. The dynamics can then be described entirely in terms of position, with the velocity eliminated from the problem. This is the overdamped regime. Every timescale considered in this work arises from the balance between the trapping force and thermal fluctuations.

This work examines two complementary questions for the same optical trap. The first concerns loading. Suppose the particle begins somewhere within a large closed region that contains the trap. How quickly does it settle into the potential well, and what fraction of the time does it eventually spend there? This is a problem of relaxation towards thermal equilibrium. The second concerns escape. Suppose the particle begins inside the trap and the surrounding boundary is removed. How long, on average, does it take to wander beyond the rim? This is the Kramers problem of thermally activated escape [4]. In this case there is no equilibrium to approach, only a gradual loss of probability from the domain. Loading and escape are governed by the same forces and differ only in their boundary conditions. The loading problem uses reflecting boundaries while the escape problem uses absorbing boundaries.

The results presented below rely on cross validation rather than any single computational approach. Every reported timescale is obtained using at least two independent methods consisting of a finite volume solver for the probability density and a particle simulation of the stochastic trajectories. In one dimension these results are also compared with an exact integral solution. Whenever an analytical solution is available, the numerical methods are validated against it. When no analytical solution exists, agreement between the two independent numerical approaches provides confidence in the results.

2 Model

Overdamped motion of the particle position $\mathbf{X}(t)$ obeys the Langevin equation

$$\mathrm{d}\mathbf{X} = \frac{1}{\gamma}\,\mathbf{F}(\mathbf{X})\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}\mathbf{W}, \qquad \mathbf{F} = -\nabla U, \tag{1}$$

with $\mathbf{W}$ a standard Wiener process. The same physics, expressed as a conservation law for the probability density $p(\mathbf{x}, t)$, is the Fokker-Planck equation [5, 6] $\partial_t p = -\nabla\cdot\mathbf{J}$ with current

$$\mathbf{J} = -\frac{1}{\gamma}(\nabla U)\,p - D\,\nabla p. \tag{2}$$

The drag coefficient is Stokes' law and the diffusion constant follows from the Einstein relation,

$$\gamma = 6\pi\eta a, \qquad D = \frac{k_\mathrm{B}T}{\gamma}, \tag{3}$$

so both transport coefficients are set once the particle radius a, the solvent viscosity η, and the temperature T are fixed. In a closed region the stationary solution of (2) is the Boltzmann density $p_\infty \propto e^{-\beta U}$ with $\beta = 1/k_\mathrm{B}T$. The well is the gradient-force potential of a focused Gaussian beam,

$$U(\mathbf{x}) = -U_0\left(\frac{w_0}{w(z)}\right)^2 \exp\!\left(-\frac{2\rho^2}{w(z)^2}\right), \qquad w(z)^2 = w_0^2\left(1 + \frac{z^2}{z_R^2}\right), \tag{4}$$

where ρ is the distance from the beam axis, z the distance along it, U_0 the depth, w_0 the beam waist, and $z_R = \pi w_0^2 n/\lambda$ the Rayleigh range over which the focus spreads. For the parameters used here, $z_R \approx 2w_0$, so the well is much stiffer across the beam than along it and its equipotentials are drawn out into a cigar shape (Figs. 1a and 1b). Close to the focus the well is approximately harmonic, with curvature $\kappa = 4U_0/w_0^2$ and a local relaxation time $\tau_\mathrm{relax} = \gamma/\kappa$. Dropping the axial coordinate gives the one-dimensional well $U(x) = -U_0\,e^{-2x^2/w_0^2}$, which keeps the transverse physics and can be handled analytically. Table 1 collects the parameters.

Symbol	Parameter	Value
a	particle radius	$1\,\mathrm{nm}$
η	water viscosity	$10^{-3}\,\mathrm{Pa\,s}$
T	temperature	$300\,\mathrm{K}$
w_0	beam waist	$500\,\mathrm{nm}$
U_0	well depth	$8\,k_\mathrm{B}T$
λ	wavelength	$1064\,\mathrm{nm}$
L	domain half-width	$2.5\,\mu\mathrm{m}$ $(= 5w_0)$
L_z	axial half-length (3D)	$3.75\,\mu\mathrm{m}$ $(= 1.5L)$
γ	Stokes drag	$1.885\times10^{-11}\,\mathrm{kg/s}$
D	diffusion constant	$2.197\times10^{-10}\,\mathrm{m^2/s}$
z_R	Rayleigh range	$982\,\mathrm{nm}$ $(\approx 2w_0)$
τ_relax	local trap time	$35.6\,\mu\mathrm{s}$
t_diff	box-crossing time	$14.2\,\mathrm{ms}$

Table 1: Physical parameters

A small constant conservative force (F_ρ, F_z) is superimposed on the optical trapping force to represent an additional external force acting on the particle. This contribution enters the effective

potential through the linear term $-F_\rho \rho - F_z z$, producing a slight tilt of the potential well and a corresponding shift in the equilibrium position. Since the additional force is conservative, the system continues to satisfy detailed balance and the stationary distribution remains the Boltzmann distribution. In the calculations presented here, the force is applied in the axial direction and performs approximately $1\, k_\mathrm{B}T$ of work across half the length of the computational domain. This perturbation introduces a controlled asymmetry while preserving an analytical expression for the stationary state. It therefore provides a convenient benchmark for validating the numerical methods in an asymmetric potential before considering non-conservative forces. A position dependent scattering force would generally violate detailed balance and remove the Boltzmann equilibrium that underpins this comparison.

The loading problem is solved on a closed domain with reflecting boundary conditions, $\mathbf{J}\cdot\hat{\mathbf{n}} = 0$, ensuring that probability is conserved and the density relaxes to its stationary distribution. The escape problem is solved on an open domain with absorbing boundary conditions, $p = 0$, allowing probability to leave the domain as particles escape from the trap. For the three-dimensional case, the computational domain is a cylinder defined by $\rho < L$ and $|z| < L_z$, where $L_z = 1.5L$ accommodates the elongated shape of the trapping potential. The cylindrical domain is embedded within a rectangular computational grid.

3 Numerical methods

Finite-volume solver. The Fokker-Planck equation is discretised using the Chang-Cooper scheme [7], a finite-volume method that evaluates the probability current (2) across each cell face using an exponentially weighted Scharfetter-Gummel interpolation [8]. The scheme conserves probability, preserves the positivity of the density in drift-dominated regions, and reproduces the Boltzmann equilibrium distribution with second-order accuracy in the grid spacing. These properties make it well suited to the optical trapping problem, where steep potential gradients occur near the trap centre. The resulting discretisation defines a generator matrix L, from which the loading and escape statistics are obtained using several complementary approaches.

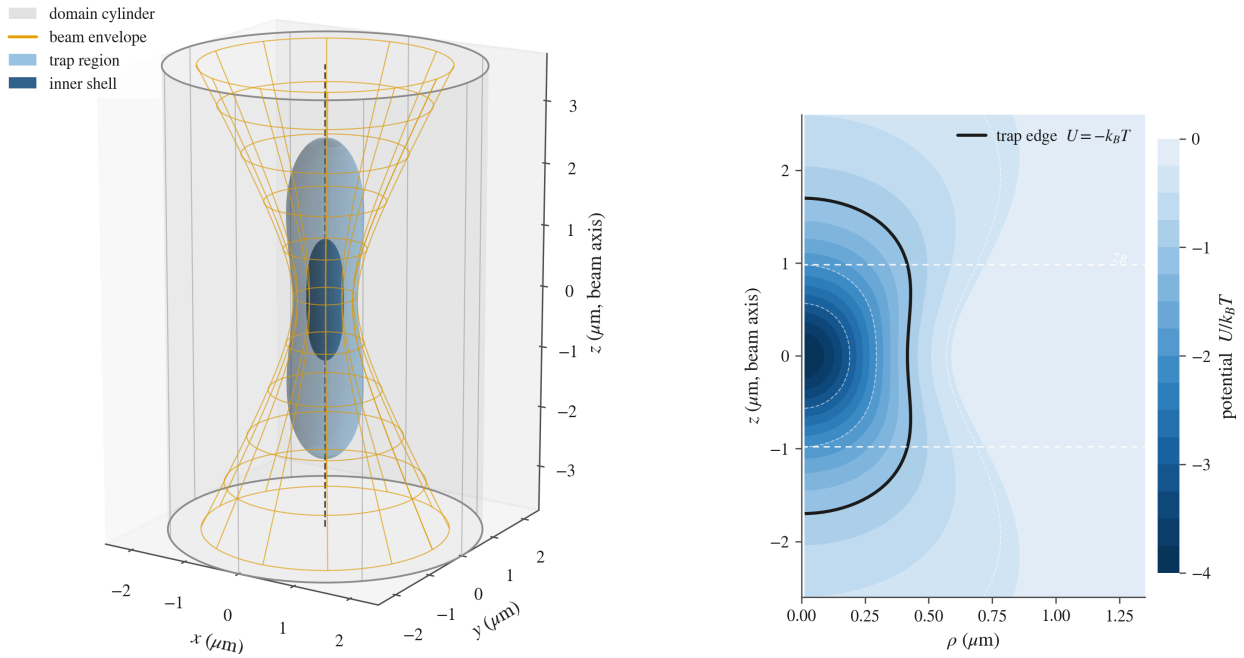
- *Loading.* The probability density is advanced in time using the Crank-Nicolson method. The loaded fraction, $N(t) = \int_R p(\mathbf{x}, t)\, \mathrm{d}\mathbf{x}$, is monitored until it reaches its steady-state value.
- *Escape from the resolvent.* The mean escape time for an initial density p_0 is $\tau = \int_0^\infty S(t)\, \mathrm{d}t$, where the survival probability is $S(t) = \mathbf{1}^\top e^{Lt} p_0$. Evaluating the time integral gives $\tau = -\mathbf{1}^\top L^{-1} p_0\, \Delta V$, so the mean escape time is obtained from a single linear solve.
- *Escape from the backward equation.* The adjoint equation $L^\dagger T = -1$ is solved with absorbing boundary conditions to obtain the mean escape time $T(\mathbf{x})$ for each initial position. Averaging T against p_0 gives the mean escape time for the chosen initial distribution.
- *Escape from the dominant eigenmode.* The eigenvalue λ_1 of L closest to zero determines the long-time decay of the survival probability. If a single slow mode dominates, as expected for a deep potential well, the mean escape time is approximated by $\tau \approx c_1/\lambda_1$, where c_1 is the projection of the initial distribution onto the corresponding eigenvector.
- *Closed-form solution in one dimension.* For the one-dimensional problem, the backward equation can be integrated analytically to give $T(x) = \frac{1}{D} \int_x^L \frac{\mathrm{d}y}{\varphi(y)} \int_{-L}^y \varphi(s)\, \mathrm{d}s$, with $\varphi = e^{-\beta U}$ and integration constants determined by the absorbing boundary conditions. This provides an exact solution against which the numerical methods are compared.

Particle simulation. The Langevin equation (1) is also solved independently using the Euler-Maruyama method [9, 10] with a timestep much smaller than the relaxation time τ_{relax}. Ensembles of 10^3 to 10^4 trajectories are simulated. Reflecting boundaries are imposed by reflecting trajectories back into the computational domain after a boundary crossing, while absorbing boundaries terminate trajectories upon their first exit from the domain. The loading curve is computed as the fraction of trajectories contained within the trapping region as a function of time. The mean escape time is obtained from the integral of the survival probability, with the unsampled tail represented by an exponential fit to the late-time decay.

This particle-based approach is entirely independent of the finite-volume solver and provides an additional validation of the numerical results. For the three-dimensional case, the particle trajectories are simulated in Cartesian coordinates using the same force field employed by the finite-volume formulation. The finite-volume solver, however, exploits axisymmetry by solving the problem on the two-dimensional (ρ, z) half-plane, substantially reducing the computational cost while preserving the same physical system. Table 2 collects the discretisation parameters. The grid spacings are small compared with the beam waist that sets the scale of the potential, and every timestep is a small fraction of the local relaxation time τ_{relax}, which is the stability-limiting scale for the explicit particle integrator.

Quantity	One dimension	Three dimensions
Grid points	1001 over $[-L, L]$	105×315 in (ρ, z)
Grid spacing	5 nm	25 nm in both directions
Crank-Nicolson step	$\tau_{\mathrm{relax}}/50 = 0.71\,\mu$s	$\tau_{\mathrm{relax}}/10 = 3.56\,\mu$s
Langevin step, loading	$\tau_{\mathrm{relax}}/40 = 0.89\,\mu$s	
Langevin step, escape	$\tau_{\mathrm{relax}}/20 = 1.78\,\mu$s	
Trajectories, loading	10^4	
Trajectories, escape	10^3	

Table 2: Discretisation parameters. The three-dimensional grid is cell-centred and carries a 5% margin beyond the cylinder, so that the domain boundary falls inside the rectangular tensor grid.



(a) The trap and its domain.

(b) The potential in the (ρ, z) plane.

Figure 1: Geometry of the three-dimensional trap. (a) The cylindrical domain (grey), the intensity envelope of the beam (amber), and two nested equipotential surfaces of the well. (b) The same potential in units of $k_{\mathrm{B}}T$, deepening to the focus, with the trap region ($U = -k_{\mathrm{B}}T$) outlined in black and the Rayleigh range z_R marked. The well is far longer along the beam than across it because $z_R \approx 2w_0$ and the slight downward shift of the deepest point is the axial tilt.

4 Results

4.1 Loading

Released from a uniform initial distribution within the closed domain, the probability density evolves towards the trapping region and the loaded fraction approaches its equilibrium value. In one dimension, the finite-volume and particle-based methods show close agreement (Fig. 2a). The characteristic loading times are 6.94 ms and 6.85 ms for the finite-volume and particle simulations, respectively, while the corresponding equilibrium captured fractions are 0.994 and 0.993. For a potential depth of $8\,k_{\mathrm{B}}T$, nearly all particles within the computational domain eventually contribute to the trapped population. The loading timescale is comparable to the diffusion time across the domain, $t_{\mathrm{diff}} = 14.2\,\mathrm{ms}$, rather than the shorter local relaxation time of the trap. This indicates that loading is primarily controlled by diffusion across the surrounding region, rather than by the local dynamics of particles already close to the potential minimum.

The three-dimensional case exhibits two important differences. The trapping region is defined by the compact equipotential surface $U < -k_{\mathrm{B}}T$, meaning that only approximately 47% of the larger cylindrical domain lies within this region at equilibrium. Consequently, the asymptotic loaded fraction in Fig. 2b is substantially below unity. The approach to equilibrium also displays two distinct stages, consisting of an initial rapid capture of particles near the focus followed by a slower diffusive transport from the outer regions of the domain.

Since the loading curve is not well described by a single exponential relaxation, the integral relaxation time,

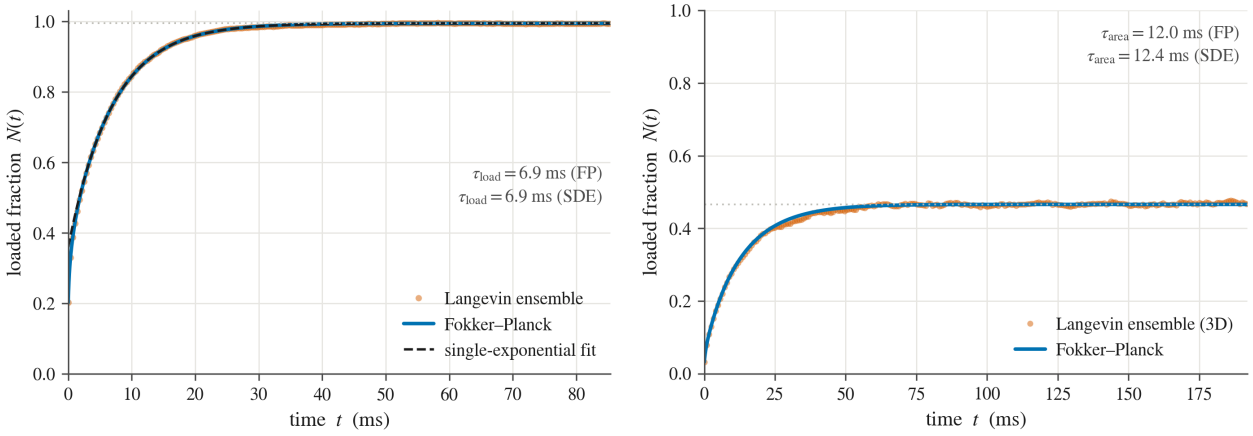
$$\tau_{\mathrm{area}} = \frac{\int (N_\infty - N)\,\mathrm{d}t}{N_\infty - N_0},$$

is used as a weighted measure of the full relaxation process. It is 12.0 ms for the finite-volume solution and 12.4 ms for the particle simulation. The finite-volume steady state agrees with the Boltzmann distribution with a relative L_1 error of 5×10^{-3}, and the two loading curves agree within the statistical uncertainty of the 10^4-particle ensemble. The corresponding evolution of the density distribution is shown in Fig. 3, where the accumulation of probability within the trap can be observed. Overall, loading occurs on the scale of tens of milliseconds for both geometries.

4.2 Escape

Escape is measured by opening the walls and letting a particle that starts in the equilibrium distribution diffuse out over the barrier. In one dimension, at a depth of $8\,k_{\mathrm{B}}T$, all five routes land on the same number to better than 0.1% (Table 3). The closed-form quadrature, the resolvent and backward-equation grid solves, the slowest-mode estimate c_1/λ_1, and the particle ensemble all give a mean escape time near 3088 ms. The coefficient $c_1 = 0.998$ confirms that a single slow mode carries almost all of the escape, and indeed the survival curve is a near-perfect exponential. The escape-time field $T(x)$ is flat across the trap and falls off toward the walls, and the quadrature and the grid solve are indistinguishable across the whole domain.

In the three-dimensional case the depth is lowered to $4\,k_{\mathrm{B}}T$ so that the particle simulation can sample enough escapes at reasonable cost, the escape rate being exponentially small in the depth. At that depth the resolvent, the time-stepped survival curve, and the particle ensemble agree to within 0.3%, all giving about 3.5 ms (Fig. 4). The asymmetric potential does not separate, so there is no closed form to check against. As a partial substitute a deep-well reference is built. When the trap is deep, escape is almost entirely along the axis, and the transverse plane can be integrated out analytically (the transverse partition function is an exponential integral), leaving a one-dimensional



(a) One-dimensional trap. **(b)** Three-dimensional case.

Figure 2: Loaded fraction $N(t)$ from the finite-volume solver (blue line) and the Langevin ensemble (orange points). (a) In one dimension the curve is a clean single exponential (black dashed fit), and the two solvers give the same time constant. (b) In the three-dimensional case the approach is two-stage and settles near 0.47, and the quoted times are the fit-free integral relaxation time τ_{area} for the finite-volume and particle results.

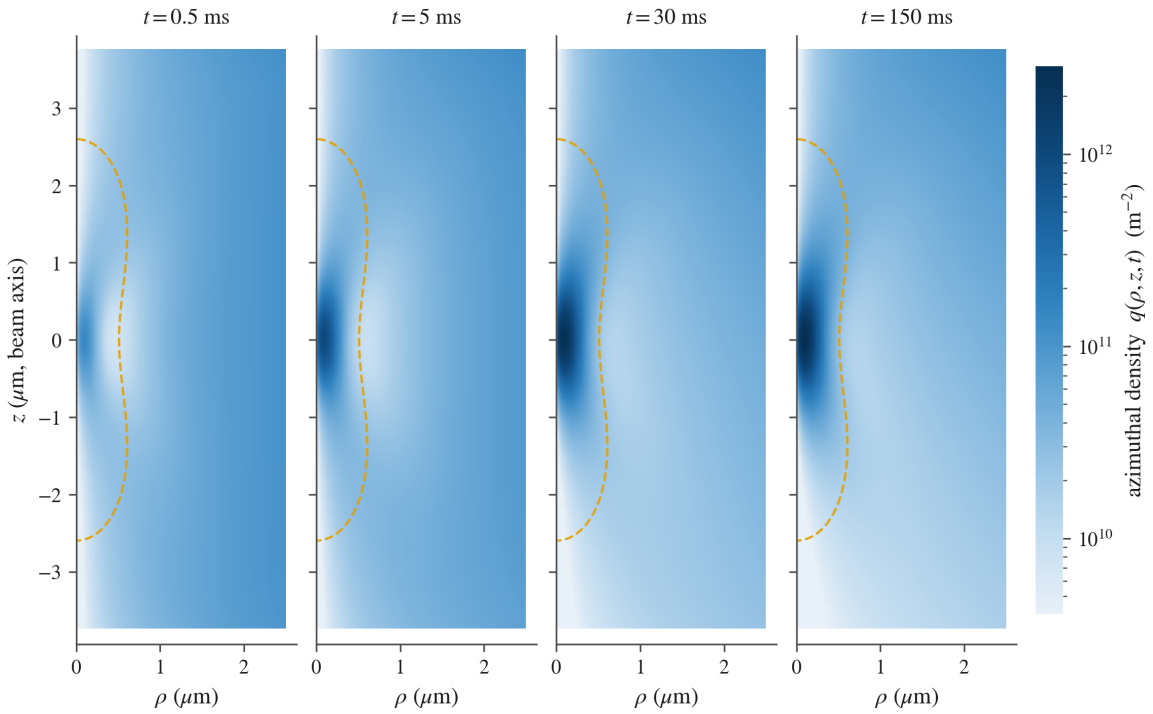


Figure 3: The density in the three-dimensional case relaxing into the well, shown as the azimuthal marginal $q(\rho, z, t) = 2\pi\rho\, p$ in the (ρ, z) half-plane at four times (note the logarithmic colour scale). The amber outline is the trap region $U = -k_{\mathrm{B}}T$. Mass gathers first near the focus and then fills the elongated well along the beam axis.

free energy along z whose escape time follows from the exact quadrature. This reference assumes the transverse motion is infinitely fast, so it overestimates, and by a controlled amount set by the ratio of transverse to axial relaxation. At a depth of $12\,k_{\mathrm{B}}T$ the full two-dimensional solve gives 537 ms and the axial reference 659 ms, higher by 23%, exactly the direction and size expected. The reference is meaningful only in the deep regime. At the shallow depth used for the particle comparison, escape has a large transverse component that the axial reduction cannot see.

Method (1D, depth $8\,k_{\mathrm{B}}T$)	τ_{exit} (ms)
Closed-form quadrature	3087.5
Grid, resolvent $-\mathbf{1}^{\top}L^{-1}p_0$	3087.8
Grid, backward equation	3087.8
Slowest mode c_1/λ_1	3087.8
Langevin ensemble	3085.3

Table 3: Mean escape time in the one-dimensional trap by five independent routes, with a spread under 0.1%.

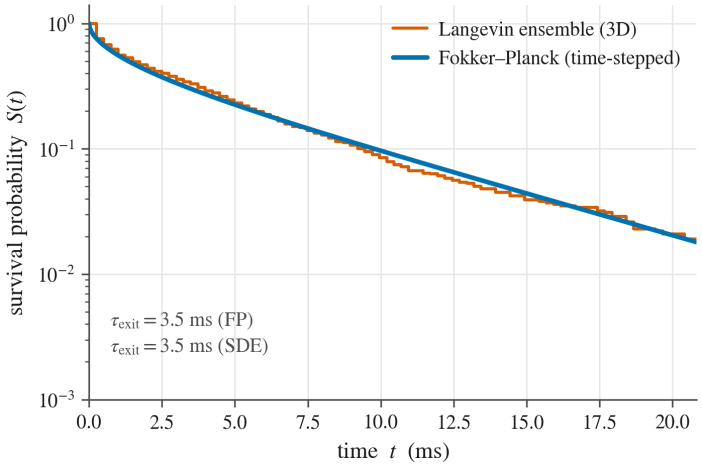


Figure 4: Escape from the three-dimensional case at depth $4\,k_{\mathrm{B}}T$. The survival probability from the time-stepped grid solve (blue) and from the Cartesian three-dimensional particle ensemble (orange steps). The two give the same mean escape time, 3.5 ms, to within sampling noise.

5 Discussion

The agreement between independent computational approaches provides the primary validation of the numerical results. The finite-volume solver, the particle-based Langevin simulation, and the analytical or spectral calculations produce consistent loading and escape times. In the three-dimensional case, where no closed-form solution exists, the agreement between the two independent numerical approaches provides the basis for validating the results. The reliability of these calculations is further supported by the one-dimensional problem, where both methods can be compared directly against an exact solution.

The parameter regime considered here remains well separated from the limits imposed by either loading or escape. At a trap depth of $8\,k_{\mathrm{B}}T$, particles load into the trap on millisecond timescales while remaining confined for timescales of several seconds. The introduction of a constant axial tilt

provides an asymmetric but still conservative perturbation of the trap. Because the tilted potential retains a Boltzmann equilibrium distribution, it allows the numerical methods to be tested in an asymmetric setting while maintaining an analytical reference state.

This provides a natural intermediate step towards more realistic optical trapping models involving non-conservative forces. In particular, the scattering force associated with the laser beam introduces a position-dependent driving component along the propagation direction. Such a force breaks detailed balance and leads to a non-equilibrium steady state with circulating probability currents, commonly referred to as a Brownian vortex [11, 12]. In this regime, the equilibrium Boltzmann distribution is no longer available as a reference, and validation must rely on agreement between independent numerical approaches. The cross-validation framework developed here therefore provides the foundation for studying these more general non-equilibrium trapping systems.

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